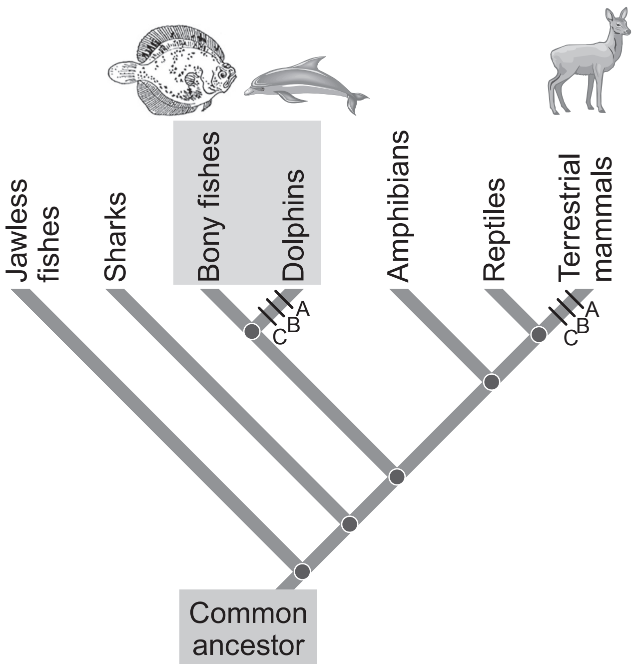
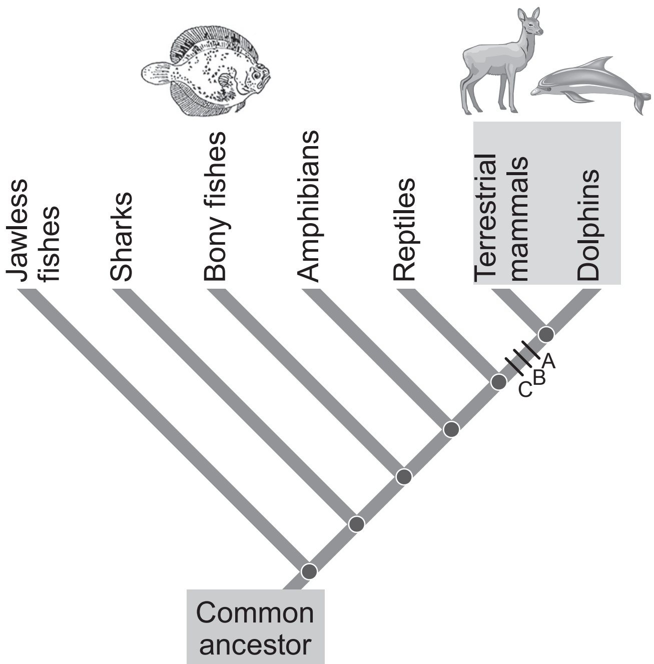
Examiner only

2. *(a)* Phylogenetic trees are used to show alternative possible evolutionary relationships between organisms. Mutation is an evolutionary event that can result in the formation of a new species. By assessing the likelihood of different mutations occurring they can be used to identify the most likely pattern of evolution.

The diagrams below show two possible phylogenetic trees for the evolution of dolphins.



Hypothesis 1: Dolphins and bony fishes are close relatives.



Hypothesis 2: Dolphins and terrestrial mammals are close relatives.

Key to the mutations required to form the new species.
A = Hair formation
B = Development of mammary glands
C = Development of middle ear bones

(i) Which phylogenetic tree represents the more probable hypothesis? Give a reason for your answer. [1]

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(ii) Explain how analysis of protein could be used to confirm your answer. [2]

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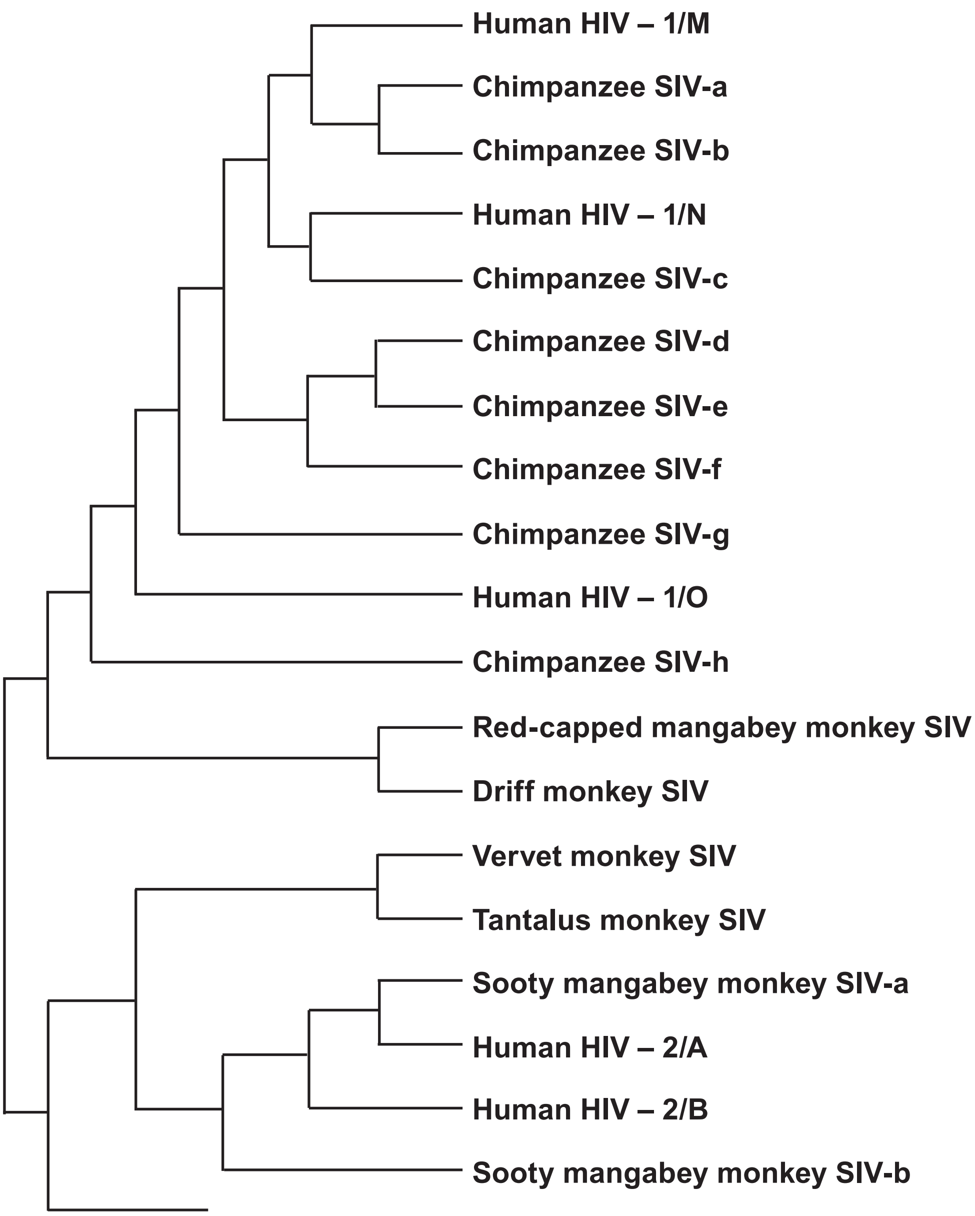
(b) AIDS was first described in humans in 1981, but the virus (HIV) which causes it, has been present in human populations since the beginning of the 19th century.

It has been found that there are two main forms of this virus: HIV1, and the more virulent HIV2. It is believed that HIV1 and HIV2 appeared due to mutations of a virus (SIV) found in monkeys and chimpanzees; these mutations have enabled them to infect humans.

Scientists have analysed the molecular similarities between HIV from infected humans and similar viruses which are found in monkeys and chimpanzees.

The diagram below shows the possible evolutionary history of HIV.



Key: SIV = Simian Immunodeficiency Virus.

HIV = Human Immunodeficiency Virus.

Examiner only

(i) What conclusions can be made from the diagram regarding the evolution of both forms of human HIV? [3]

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(ii) Scientists have been able to calculate the mean rate of mutation in the virus which causes HIV. What use can scientists make of this information? [1]

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(iii) Evidence shows a high similarity between one strain of HIV2 and a virus found in Sooty Mangabey monkeys. It was unclear if the virus had originally evolved in humans or in the monkeys. Further analysis showed that there was more genetic variation in the monkey virus than in the human virus.

Explain why it is now believed that the virus originated in the **Sooty Mangabey monkey**. [1]

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